



Sino Bright School Chongqing

Monthly Test, 2025-2026 T2

Class & Subject: G10 A2 Math

Date: Mar 2026

Time: 90 Minutes

Teacher: Shi Feng

IMPORTANT: Read together before beginning the exam.

Students must complete the assessments to the best of their ability. Any students found cheating during an exam will be given a zero and referred to the Principal for further disciplinary measures. Cheating is a serious offense.

“Cheating” may include any of the following:

- Having course papers, notes, unapproved data sheets or other course-related materials in the exam room without the expressed permission of the subject class teacher in charge of the exam.
- Using or even having a cellphone or other electronic device in the exam room without the express permission of the subject class teacher in charge of the exam.
- Attempting to communicate, either verbally or non-verbally with other students during the exam.
- Having exam-related materials prior to the writing of the exam.

Mark Breakdown:	Student Name:
Part I: Multiple Choice Questions ×	
Part II: Short Answer Questions ×	
Part III: Long Answer Questions 76 Marks	
Total Marks: 76 Marks	Score: _____

Special Instructions/Materials Allowed:

- Pen, Calculator

Questions

1. (i) A student states: “If x and y are irrational numbers, $x \neq y$, then xy is also irrational.” Show, by counter example, that this statement is not always true. **(1 marks)**
- (ii) Prove, using algebra, that for all odd integers n , the value of the expression

$$n^3 + 3n + 2$$

is always even but never a multiple of 4.

(4 marks)

2. (a) Find the first 4 terms, in ascending powers of x , of the binomial expansion of

$$(1 - 4x)^7$$

giving each term in simplest form.

(4 marks)

- (b) In the series expansion of

$$(5 + kx)(1 - 4x)^7$$

where k is a constant, the coefficient of the term in x^2 is 1316. Use the answer to part (a) to find the value of k .

(2 marks)

3. (a) Show that the equation

$$8 \tan \theta = 3 \cos \theta$$

may be rewritten in the form

$$3 \sin^2 \theta + 8 \sin \theta - 3 = 0.$$

(3 marks)

- (b) Hence solve, for $0 \leq x \leq 90^\circ$, the equation

$$8 \tan 2x = 3 \cos 2x,$$

giving your answers to 2 decimal places.

(3 marks)

4. The circle C has equation

$$x^2 + y^2 + 6x - 4y - 14 = 0.$$

(a) Find

- (i) the coordinates of the centre of C ,
- (ii) the exact radius of C .

(3 marks)

(b) The line with equation $y = k$, where k is a constant, is a tangent to C . Find the possible values of k .

(2 marks)

(c) The line with equation $y = p$, where p is a negative constant, is a chord of C . Given that the length of this chord is 4 units, find the value of p .

(2 marks)



5. (i) An arithmetic series has first term a and common difference d . Prove that the sum to n terms of this series is

$$\frac{n}{2}\{2a + (n - 1)d\}.$$

(4 marks)

- (ii) A sequence $u_1, u_2, u_3, \dots$ is given by

$$u_n = 5n + 3(-1)^n.$$

Find the value of

- (a) u_5 ,
(b) $\sum_{n=1}^{59} u_n$.

(6 marks)

6. (a) Given that

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25),$$

show that

$$2x^3 - 3x^2 - 30x + 56 = 0.$$

(5 marks)

- (b) Hence, using algebra and showing each step of your working, solve

$$3 \log_3(2x - 1) = 2 + \log_3(14x - 25).$$

(5 marks)

7. Kim starts working for a company.

- In year 1 her annual salary will be £16200.
- In year 10 her annual salary is predicted to be £31500.

Model A assumes that her annual salary will increase by the same amount each year.

(a) According to model A, determine Kim's annual salary in year 2. **(3 marks)**
Model B assumes that her annual salary will increase by the same percentage each year.

(b) According to model B, determine Kim's annual salary in year 2. Give your answer to the nearest £10. **(3 marks)**

(c) Calculate, according to the two models, the difference between the total amounts that Kim is predicted to earn from year 1 to year 10 inclusive. Give your answer to the nearest £10. **(4 marks)**

8. The function $f(x)$ is defined by

$$f(x) = ax^3 + bx^2 + 5x - 3,$$

where a and b are constants. Given that $(x + 3)$ is a factor of $f(x)$,

(a) show that

$$b = 3a + 2.$$

(3 marks)

Given that when $f(x)$ is divided by $(2x - 1)$, the remainder is $\frac{7}{4}$.

(b) find the value of a and the value of b .

(3 marks)

(c) Using algebra, find the quotient and the remainder when $f(x)$ is divided by $(x - 2)$.
(4 marks)

9.

$$f(x) = kx^3 - 15x^2 - 32x - 12, \quad \text{where } k \text{ is a constant.}$$

Given that $(x - 3)$ is a factor of $f(x)$,

(a) Show that $k = 9$. **(4 marks)**

(b) Using algebra and showing each step of your working, fully factorise $f(x)$. **(5 marks)**

(c) Solve, for $0 \leq \theta < 360^\circ$, the equation

$$9 \cos^3 \theta - 15 \cos^2 \theta - 32 \cos \theta - 12 = 0,$$

giving your answers to one decimal place. **(3 marks)**